Anthony Coffey

Senior Product Engineer · AI in Regulated Healthcare
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SUMMARY

Senior software engineer with 12+ years of experience shipping full-stack products, specializing in production-ready AI features: LLM agents with live tools, RAG over private data, and OCR pipelines. Recent work includes a PHI-redacting OCR service for an FDA-regulated ultrasound AI platform, an end-to-end migration of 200+ tests from Cypress to Playwright, and the system architecture for a Gemini-powered submission review system approved by PepsiCo's CISO. I work spec-first using a test-driven workflow with coding agents like Claude Code, Codex, and GitHub Copilot.

CORE SKILLS

- **AI features:** LLM agents & tool use (MCP) · RAG (Qdrant, pgvector) · fine-tuning · structured outputs · Anthropic, OpenAI and Gemini APIs · LiteLLM · OCR/vision (EasyOCR, Cloud Vision)
- **Frontend:** React 19 · TypeScript · Next.js · Vite · Tailwind / shadcn/ui · visx / Highcharts · React Native / Expo · Capacitor · Three.js (R3F) · i18n
- **Backend:** Python (FastAPI) · Node/Express · PostgreSQL · Redis/BullMQ · Firebase Supabase · Stripe · Twilio
- **Quality & Delivery:** Vitest · Playwright · pytest · Cypress · GitHub Actions · Docker Kubernetes · AWS · GCP · Cloudflare · Vercel · Git
- **Domain:** HIPAA/SOC 2 product requirements · FDA 510(k) / SaMD · multi-tenant SaaS

EXPERIENCE

Full Stack Software Engineer – Bitmotive, LLC

Feb 2026 – Aug 2026 · Contract

- Turned static matplotlib model-validation images into **interactive analytics** (visx zoom, brush and tooltips) for Bland-Altman, scatter and error-distribution charts. Added metric cards (MAE, RMSE, R^2, normality) and **one-click PDF reports** for clinical and regulatory teams.
- Migrated E2E testing from Cypress to **Playwright (200+ tests),** configured to run against production build in CI to prevent regression.
- Built the backend of an **OCR pipeline that reads burned-in text** on ultrasound frames. It recovers estimated delivery dates the product was missing and **redacts patient identifiers** before images are stored.
- Shipped end-to-end **MFA** (setup wizard, challenge flow, recovery codes, institution-level enforcement) and a **searchable audit-log viewer** with CSV export for global admins.
- Localized the frontend (i18next) and FastAPI backend (Babel) in **English, Spanish and Portuguese**.

Lead Software Engineer – 24hrcarunlocking.com

Aug 2023 – May 2026 · Full-time

● Rebuilt the WordPress marketing site on Astro + **Sanity** + Cloudflare (**~$6,700/yr saved**).

● Retrofitted mission critical API endpoints with **Playwright** testing to ensure system-wide reliability and catch breaking changes in CI/CD before code merges.

● Shipped **RescueBot**, an in-app scheduling assistant on a **fine-tuned GPT-4o-mini** model that collects service details conversationally, quotes from live pricing, and hands off to a human on request.

● Architected **AmeniLink**, a multi-tenant property-management SaaS for web, iOS and Android (React 19, TypeScript, MUI, Firebase, Capacitor) with invitations, RBAC and service requests.

● Maintained the dispatcher portal (Nuxt/Vuetify), technician mobile app (React Native/Expo) and Node API, including **real-time auto-dispatch** for ~14,000 calls and ~15,000 SMS a month, featuring push/SMS notifications, Stripe payments and lead intake webhooks.

Staff Engineer – Southern Made

Apr 2024 – Sep 2024 · Contract

● Designed the architecture for the **DORITOS® Crash the Super Bowl** submission platform on GCP, with Gemini and Cloud Vision automatically reviewing user-generated entries. **The design was reviewed and approved by PepsiCo's CISO (Chief Information Security Officer)**.

Full Stack Software Engineer – Marani Health

Mar 2022 – Apr 2024 · Full-time

● Built a React annotation app (AWS Amplify) where physicians labeled anonymized fetal and maternal heart-rate data to validate the algorithm behind an **FDA 510(k) submission**.

● Refactored legacy Python signal-analysis code for a Bluetooth fetal/maternal monitor into testable units, establishing a comprehensive **Playwright** test suite **(0% to 99% coverage)** required for **FDA 510(k) submission** compliance.

Cybersecurity Analyst – Health Tech Defenders

Dec 2023 – Sep 2025 · Contract

● Penetration-tested healthcare web apps and APIs and delivered remediation guidance for HIPAA-regulated environments.

Earlier Experience · 2014 – 2022

● **React Native Developer, festivalPass** (2021–2022): Developed cross-platform React Native app at full parity with web, backed by Node/Strapi; owned store releases.

● **Freelance Engineer, Lifecycle Insights** (2019–2021): Vue/Node/MongoDB survey app with live data visualization and styled PDF reports.

● **Freelance Engineer, Smart Insurance** (2016–2019): React workers-comp quoting app with multi-state underwriting rules fully compliant with state regulations.

● **Full Stack Developer, RiskPro®** (2014–2016): Developed Highcharts visualization layer in JavaScript client app and refined risk algorithms on a Python backend for a fintech platform on AWS/Lambda.

PROJECTS

● **coffey.codes:** Next.js 16 portfolio blog with a React Three Fiber 3D homepage, MDX blog (23 articles), **Playwright** test suite that runs in CI/CD against production environment before PRs can merge.

● **Bugsy - Personal AI agent:** RAG over my repos and docs, live Jira, GitHub and Notion tools through MCP, an LLM-scored job board, and an issue-to-PR coding agent, all hosted on Google Cloud Platform.

● **Support Triage Agent:** Claude agent that categorizes, prioritizes and escalates support tickets with cited help articles. Built spec-first with AGENTS.md/CLAUDE.md steering, Claude Code skills and hooks, and an eval harness with a CI release gate.

● **Simply Voice:** text-to-speech SaaS (React, Firebase, Google Cloud TTS) with Stripe subscriptions and usage quotas. simply-voice-452800.web.app

● **Flutter LabelScan:** mobile price-label scanner using Cloud Vision + Gemini to extract prices and keep a running cart total.

EDUCATION

Gulf Coast State College - Associate of Arts · Graduated May 2012